Advanced Industrial Organisation / Project 1

The European Car Market

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Use of AI tools. I used ChatGPT (OpenAI) for code suggestions, numerical checks and first drafts of paragraphs based on my prompts. It also assisted with LaTeX formatting. I remain responsible for the final text, code and results, and for understanding the models used.

1 Empirical part

1.1 Demand estimation

The data cover 40 cars in each of 150 country–year markets in Belgium, France, Germany, Italy and the UK, 1970–1999. I define market size as $M_t = \text{pop}_t/2.5$, approximating one annual purchase opportunity per household. Thus $s_{jt} = q_{jt}/M_t$ and $s_{0t} = 1 - \sum_j s_{jt}$. The outside option includes not purchasing a new car, buying used and buying an excluded model. Household size is an assumption; a larger M_t increases outside diversion and can reduce estimated merger harm. The changing top-40 sample also means excluded cars are imperfectly represented by one outside option.

In the simple logit model, utility of good j in market t for household i is given by

$$u_{ijt} = \underbrace{\mathbf{x}'_{jt}\boldsymbol{\beta} + \lambda_{m(t)} + \gamma_{b(j)} - \alpha \log p^C_{jt} + \xi_{jt}}_{\equiv \delta_{jt}} + \epsilon_{ijt}, \quad \epsilon_{ijt} \overset{iid}{\sim} T1EV(0). \tag{1.1}$$

The outside option's deterministic utility is normalised to zero. This gives the choice probabilities and demand inversion

$$s_{jt} = \frac{e^{\delta_{jt}}}{1 + \sum_{k \in \mathcal{J}_t} e^{\delta_{kt}}} \quad \Longrightarrow \quad \delta_{jt} = \log s_{jt} - \log s_{0t}. \tag{1.2}$$

Observed shares are treated as these population probabilities. I estimate (1.2) by OLS and IV, with standard errors clustered by car model code.

Prices p^C are `princ`: VAT-inclusive local prices divided by nominal GDP per capita. Utility uses their logarithm, so a one-percent price increase changes utility by $-\alpha \log(1.01)$. This imposes common tastes for income-relative prices and does not model household incomes explicitly. Controls are weight in tonnes, fuel consumption, $hp/100$ and a home-product dummy, together with brand and country fixed effects. All non-price controls remain in levels; brand is categorical.

Country effects absorb persistent country demand differences; brand effects absorb permanent brand appeal. Price variation remains across models within brand–country cells and as relative brand positions change over time. Unobserved model quality, changing brand appeal and common time trends can still correlate with prices.

The simple logit estimation has two major problems. Firstly, the IIA assumption can be a poor representation of real-world substitution patterns and thus affect the merger simulation. One way to relax IIA is to introduce correlated match values within the five car classes: Subcompact, Compact, Intermediate, Standard and Luxury. The outside

option is its own nest. Writing $g(j)$ for j's class, nested logit gives

$$D_{gt} = \sum_{k \in g} \exp\{\delta_{kt}/(1-\rho)\}, \qquad s_{jt} = \frac{\exp\{\delta_{jt}/(1-\rho)\} D_{g(j)t}^{-\rho}}{1 + \sum_g D_{gt}^{1-\rho}}, \quad 0 \le \rho < 1.$$

At $\rho = 0$ this reduces to simple logit. Nesting relaxes global IIA, while retaining proportional substitution conditional on class. Price sensitivity remains homogeneous. Demand inversion gives one linear regression:

$$\log s_{jt} - \log s_{0t} = \mathbf{x}_{jt}'\boldsymbol{\beta} + \lambda_{m(t)} + \gamma_{b(j)} - \alpha \log p_{jt}^{C} + \rho \log s_{j|g,t} + \xi_{jt}, \tag{1.3}$$

where $s_{j|g,t} = s_{jt} / \sum_{k \in g(j)} s_{kt}$. I estimate both models by OLS and 2SLS, keeping controls and fixed effects identical.

The second problem is price endogeneity: unobserved quality may raise both prices and demand. In nested logit, within-class share also responds to unobserved quality. IV therefore instruments *both* log price and log within-class share. I use the same two excluded instruments in simple and nested IV, both of the same type: product counts. Let $\mathcal{R}_{jt}$ contain cars in j's class produced by other manufacturers, and $\mathcal{O}_{jt}$ the other cars in that class made by j's manufacturer. The instruments are

$$z_{jt}^{R} = |\mathcal{R}_{jt}|, \qquad z_{jt}^{O} = |\mathcal{O}_{jt}|.$$

The car itself is excluded. Each column is divided by its sample standard deviation. Counts vary across manufacturer–class portfolios and across country–years; these differences survive additive country and brand effects. They shift the competitive environment and within-class choice set, affecting prices and conditional shares. Identification assumes that assortment and included characteristics are exogenous to unobserved demand. Strategic positioning and selection into the top 40 can violate this assumption.

I screened count-only and differentiation-only sets of two to four instruments, choosing this minimal count pair for clarity and admissible estimates in both models. After effect absorption, six instrument columns identify five simple-logit or six nested-logit coefficients. Nested IV is exactly identified, so there is no overidentification test. Table 1 reports one clustered first-stage diagnostic for each endogenous variable. Relevance alone neither establishes exclusion nor guarantees strong conditional identification. The exploratory search is documented in the code.

Price sensitivity increases from OLS to IV (Table 1). This is consistent with unobserved quality raising both demand and prices, causing OLS to understate the negative price effect. IV improves this comparison only if the instruments are valid. Nested OLS estimates substantial within-class dependence. Nested IV gives a smaller, imprecise

nesting coefficient, so its departure from simple logit is uncertain.

Home-product coefficients remain positive, suggesting sizeable home bias after observed characteristics and fixed effects. They need not measure a pure causal preference: domestic distribution networks or unobserved country-specific quality can remain. Power changes sign between OLS and IV; fuel consumption turns positive in nested IV. Their imprecise estimates discourage strong conclusions about individual characteristics.

1.2 Merger analysis

A given firm / car manufacturer f in market t, with product portfolio $\mathcal{J}_{ft}$, maximises $\sum_{k\in\mathcal{J}_{ft}} M_t(p_{kt} - c_{kt})s_{kt}(\mathbf{p}_t)$. Here $p = p^C/(1 + tax)$ is producer price; taxes are held fixed. Stacking the FOCs for all firms, with $G_{jk} = \partial s_k/\partial p_j$, gives

$$\mathbf{s}_t(\mathbf{p}_t^*) = -[\mathcal{H}_t \odot G_t(\mathbf{p}_t^*)](\mathbf{p}_t^* - \mathbf{c}_t), \qquad \widehat{\mathbf{c}}_t = \mathbf{p}_t^0 + [\mathcal{H}_t^0 \odot G_t(\mathbf{p}_t^0)]^{-1}\mathbf{s}_t^0. \tag{1.4}$$

The ownership matrix has $\mathcal{H}_{jk} = 1$ if j and k have the same owner, and zero otherwise. Firms compete on price and all optimise given the other firms' prices. These inferred costs depend on static multiproduct Bertrand conduct and constant marginal costs.

To simulate VW–Fiat, I link their products in $\mathcal{H}_t^1$, retain these costs and let all firms reoptimise. I use PyBLP's Morrow–Skerlos iteration for every market and all four models. All 600 baseline cases have positive costs and pass independent pricing and profit checks.

The diversion ratio is the fraction of sales lost by j following a small own-price increase that moves to k:

$$D_{j\to k,t} = -\frac{\partial s_{kt}/\partial p_{jt}}{\partial s_{jt}/\partial p_{jt}}, \qquad E_{jk,t} = \frac{\partial s_{jt}}{\partial p_{kt}}\frac{p_{kt}}{s_{jt}}.$$

For simple logit, $D_{j\to k,t} = s_{kt}/(1 - s_{jt})$, so OLS and IV have identical diversion at observed shares. Nested logit allows stronger within-class substitution. I sum diversion from each VW car to Fiat cars, and conversely, weighting source cars by premerger sales within market, then averaging equally over markets. Table 2 reports median own elasticities and mean cross elasticities between distinct cars in the same class.

A regulator's real interest is how a possible merger would affect consumer wellbeing. For scenarios $h = 0, 1$, nested logit's expected logsum is $L_t^h = \log[1 + \sum_g (D_{gt}^h)^{1-\rho}]$. The common Euler constant cancels. With log prices, dividing $L_t^1 - L_t^0$ by α does not give monetary CS: marginal disutility of spending is α/p^C, not constant. Instead I report the

uniform increase in all baseline car prices giving the same utility loss:

$$W_t = 100 \left[\left(\frac{A_t^0}{A_t^1} \right)^{1/\alpha} - 1 \right], \qquad A_t^h = e^{L_t^h} - 1, \qquad \overline{W} = \frac{1}{T} \sum_t W_t.$$

Positive values indicate harm. Entry, quality changes, replacement dynamics and income effects are excluded.

IV reduces estimated harm by raising price sensitivity. Nested IV increases partner diversion, strengthening recapture despite its larger α. Nested OLS also raises diversion, but makes own demand more elastic: the inferred merging-firm margin falls substantially relative to simple OLS. Its price effect is consequently smaller. Greater diversion alone does not determine the final merger effect. Figure 1 compares countries; Italy has the largest combined effect in all four models.

I reduce merging firms' marginal costs to $(1 - \kappa)c_{jt}$, leaving rivals' costs unchanged, and resolve prices. A root search finds the saving restoring expected utility in each market; their maximum is one common saving sufficient for all 150 markets (Table 2). Strict gains require exceeding it. This protects expected welfare in each market, not every household. Without efficiencies, all models imply harm. I would not approve without verified savings above the relevant threshold, while recognising the uncertainty across models.

Population/2 re-estimates all four models and produces prices in all 150 markets (Table 3). Population/3 gives one negative simple-OLS cost, so I do not report an OLS merger average for that specification. Year effects invalidate simple-OLS cost recovery, preventing a full comparison. Omitted time trends remain a limitation. Instrument checks add a third count (rivals in other classes) or replace both counts with local differentiation instruments; details are in the code. Instrument exclusion, exploratory selection, the fragile OLS price coefficient and conduct remain limitations.

2 Theoretical part

2.1 Price dispersion for homogeneous goods

The classic price-dispersion models considered can all be slightly augmented to explain the qualitative pattern of home bias. For example, search models with homogeneous goods, where we exogenously set the information cost to favour home goods in some way.

Take for example Diamond (1971), where a unit continuum of consumers value the homogeneous good at $v > c$, where c is the firms' constant marginal cost. Consumers first search at a randomly selected firm at zero cost. Then they choose whether or not

to search another random firm, incurring a search cost $s > 0$. The unique symmetric pure-strategy Nash equilibrium (PSNE) is that firms choose monopoly prices $p^* = v$ and consumers only search the first firm assigned to them. Intuitively, all other potential equilibria with $p^* < v$ incentivise a firm deviation of

$$p' = \min\{p^* + s - \varepsilon, v\}, \qquad 0 < \varepsilon < s.$$

To see this, let a consumer i search the deviating firm with price p' at zero cost. They can either buy this good at cost $p' \leq p^* + s - \varepsilon$ or search and buy a non-deviating firm's good at total cost $p^* + s$. Since goods are homogeneous they strictly prefer buying at p' to searching and buying at p^*.

Consider Diamond (1971) slightly augmented so the initial zero-cost search is drawn from a distribution that assigns greater probability to domestic firms while remaining uniform within the domestic and foreign groups. If no domestic firms are available, the assignment is uniform across all firms. This would induce an equilibrium with home bias as observed in the data.

A similar home-bias mechanism could be introduced in Burdett and Judd by biasing access to offers towards domestic firms. Much the same analysis could probably be done for models without search like Varian (1980)—by exogenously biasing the uninformed / loyal customers towards home production—or Butters (1977)—by setting advertising costs to be lower for firms in their home country.

2.2 Perloff–Salop and logit

In the Perloff–Salop model consumers have utility given by

$$u_{ij} = \epsilon_{ij} - p_j, \tag{2.1}$$

where $\epsilon_{ij} \sim F$ are independent and identically distributed stochastic match values, interpreted as how well product j fits consumer i's preferences.

The distribution F is arbitrary and—crucially for this discussion—does not depend on observable characteristics of j. Therefore, product differentiation comes from the random match values, rather than observed characteristics.

Looking back at the simple logit, we see that utility does depend on j's characteristics through $\mathbf{x}_{jt}'\boldsymbol{\beta}$. This is the main difference; the empirical models also use log prices and a finite outside option. In (1.1) the match-value distribution is chosen as the Type I extreme value distribution (with location zero). Nested logit additionally correlates match values within each class, while retaining common price sensitivity.

2.3 Testing the Perloff–Salop model

The distribution F in the Perloff–Salop model is by no means irrelevant (as I will show in Section 2.4), and therefore whenever empirical models of the form (2.1) are tested, the distributional choice of T1EV(0) is part of the joint hypothesis.

In addition to this, as always, exogeneity of the regressors must hold; the regression test cannot establish this assumption.

Having this in mind, we test the following null hypothesis:

$$\mathcal{H}_0 : \boldsymbol{\beta} = \mathbf{0},\ \boldsymbol{\gamma} = \mathbf{0}, \qquad \mathcal{H}_A : \text{at least one characteristic or brand effect is nonzero.}$$

Here the null includes the four characteristics and brand effects; for this linear test, market effects are retained, allowing common market demand shifts and a finite outside option. Rejecting it would imply that observable characteristics affect utility, contradicting this adjusted version of (2.1). Estimating (1.2) and using a cluster-robust F-test yields $F(39, 284) = 53.005$ and $p < 0.001$. Thus we firmly reject the joint null, rather than the entire family of models with arbitrary F.

2.4 Many car manufacturers

Using the result by Gabaix, Laibson and Resnick (2016), markups in the Perloff–Salop model have the asymptotic property, as the number of firms $n \to \infty$,

$$p^* - c \propto \frac{1}{nf[F^{-1}(1-1/n)]}. \tag{2.2}$$

Here f is the density of the i.i.d. taste shocks and F their cdf. Inserting the two distributions respectively and letting $n \to \infty$ informs us about the markup with many car manufacturers.

2.4.1 T1EV shocks

Here I restate the distribution for clarity:

$$F(x) = \exp\{-\exp\{-x\}\}, \qquad f(x) = \exp\{-x\}F(x).$$

Inserting into the denominator of (2.2) yields

$$\begin{aligned} nf[F^{-1}(1-1/n)] &= n\exp\{-F^{-1}(1-1/n)\}(1-1/n) \\ &= -n\log(1-1/n)(1-1/n) \longrightarrow 1. \end{aligned}$$

Taking the limit uses $\lim_{n\to\infty}(1-1/n)^n = \exp\{-1\}$. That is, when taste shocks follow a T1EV(0) distribution, increasing the number of firms indefinitely does not remove the markups.

2.4.2 Gaussian shocks

For Gaussian shocks, let $F = \Phi$ and $f = \phi$. Set $x = \Phi^{-1}(1-1/n)$, so $n = 1/[1-\Phi(x)]$. As $n \to \infty$, we have $x \to \infty$.

Using l'Hôpital's rule and $\phi'(x) = -x\phi(x)$,

$$\lim_{n\to\infty} n\phi[\Phi^{-1}(1-1/n)] = \lim_{x\to\infty} \frac{\phi(x)}{1-\Phi(x)} = \lim_{x\to\infty} \frac{-x\phi(x)}{-\phi(x)} = \infty.$$

The denominator in (2.2) therefore grows without bound. Its reciprocal approaches zero, so markups vanish and prices approach marginal cost, unlike the T1EV case.

The calculation above uses the lecture's level-price utility. In the empirical log-price model, symmetric Gumbel shares $1/n$ instead give $(p-c)/p = 1/[\alpha(1-1/n)]$, so $p/c \to \alpha/(\alpha-1)$ for $\alpha > 1$. Gaussian shocks give a vanishing Lerner index. Thus the contrast survives, but the price transformation changes the markup's units.

2.4.3 Implications for empirical estimates

The above results highlight that the assumed taste-shock distribution is by no means innocent in a Perloff–Salop model. This hints at a more general insight. The assumptions made to derive logit (for example the T1EV assumption on taste shocks) are not innocent and can themselves cause certain regularities of the models. These regularities or outcomes are not findings but assumptions and should be stated as such.

Another example could be the choice of nests in a nested logit model: putting cars in the same class makes them closer substitutes by assumption. This does not mechanically increase estimated merger harm, because price sensitivity and recovered margins also change. Here we again see that the modelling decisions before estimation partly determine the results.

Sources. Supplied Project 1; Munk-Nielsen, *Demand Estimation* (2026), Sections 1–5; Vikander, Lectures 1, 5–6 (2026); Goldberg–Verboven (2001) data.

Output: estimates and merger results

Table 1: Demand estimates: log prices and nests on class

	Simple OLS	Simple IV	Nested OLS	Nested IV
Price α	1.145 (0.139)	2.393 (0.628)	1.163 (0.104)	4.114 (1.553)
Nesting ρ	—	—	0.658 (0.026)	0.168 (0.140)
Weight	1.300 (0.254)	1.863 (0.402)	0.175 (0.162)	2.351 (0.629)
Fuel consumption	−0.101 (0.019)	−0.031 (0.039)	−0.072 (0.013)	0.072 (0.090)
Power	−0.322 (0.250)	0.492 (0.498)	−0.016 (0.185)	1.689 (1.102)
Home product	1.238 (0.050)	1.253 (0.048)	0.544 (0.063)	1.096 (0.148)
First stage: price	—	12.2	—	12.2
First stage: within share	—	—	—	136.4

Note: 6,000 observations, 150 markets. Model-clustered SEs in parentheses (285 clusters), without finite-sample correction. All four models include additive brand and country effects, weight/1000, power/100, fuel and home. Only price is logged. Both IV models use the same two count instruments. First-stage entries are clustered Wald statistics divided by two; they are marginal diagnostics, not conditional identification tests. Nested IV instruments both price and within-class share.

Table 2: VW–Fiat merger: all four models cover all 150 markets

	Simple OLS	Simple IV	Nested OLS	Nested IV
Combined price increase (%)	4.31	0.45	3.31	0.51
Median own elasticity	−1.14	−2.39	−3.23	−4.88
Mean cross: within class	0.0022	0.0046	0.2083	0.0844
Partner diversion (%)	0.56	0.56	5.66	1.77
Welfare price equivalent (%)	1.52	0.17	1.29	0.19
Required cost saving (%)	27.87	1.96	21.73	1.90

Note: Prices and partner diversion use premerger unit-sales weights within markets, then equal market weights. Welfare is a uniform-price equivalent, not money CS or a percentage of CS. Savings reduce only VW–Fiat marginal costs and restore expected welfare in every market; strict gains require exceeding the threshold. Full precision is retained in the code.

Output: country comparison and market-size robustness



Figure 1: Merger price effects and partner diversion by country

Table 3: Market-size check: population/2 versus population/2.5

	Combined price increase (%)	
Model	Baseline	Pop./2
Simple OLS	4.31	3.56
Simple IV	0.45	0.36
Nested OLS	3.31	3.30
Nested IV	0.51	0.47

Note: All models are re-estimated; shares, inferred costs and mergers are recomputed in all 150 markets. The top panel reports the merging portfolio's sales-weighted price change; the bottom panel reports lost sales diverted to the partner. Markets receive equal weights.